

Q1) AR(3):

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \phi_3 X_{t-3} + a_t, \quad a_t \sim N(0, \sigma_a^2) \text{ iid.}$$

obs data:

$$Y_t = X_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2).$$

$$\Rightarrow \underline{X}_t = \begin{bmatrix} X_t \\ X_{t-1} \\ X_{t-2} \\ X_{t-3} \end{bmatrix}, \quad \underline{G}_t = \begin{bmatrix} \phi_1 & \phi_2 & \phi_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \underline{W}_t = \begin{bmatrix} a_t \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \underline{V}_t = \begin{bmatrix} \sigma_a^2 & 0 & 0 \\ 0 & \sigma_\varepsilon^2 & 0 \\ 0 & 0 & \sigma_\varepsilon^2 \end{bmatrix}$$

$$\underline{X}_t = \underline{G}_t \underline{X}_{t-1} + \underline{W}_t = \begin{bmatrix} \phi_1 & \phi_2 & \phi_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_{t-1} \\ X_{t-2} \\ X_{t-3} \end{bmatrix} + \begin{bmatrix} a_t \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow Y_t = X_t + \varepsilon_t = \underline{F}_t^T \underline{X}_t + V_t$$

$$\therefore \underline{F}_t = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}, \quad V_t = \varepsilon_t, \quad V_t = \sigma_\varepsilon^2$$

Q2) $Y_t = \underline{F}_t^T \underline{X}_t + V_t, \quad X_t = G_t X_{t-1} + W_t$

$$\underline{X}_t = \begin{bmatrix} d_{1t} & B_{1t} & d_{2t} & B_{2t} \end{bmatrix}^T, \quad Y_t = \begin{bmatrix} r_{1t} \\ r_{2t} \end{bmatrix}$$

$$\underline{X}_t = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} d_{1t-1} \\ B_{1t-1} \\ d_{2t-1} \\ B_{2t-1} \end{bmatrix} + \begin{bmatrix} y_{1t} \\ y_{2t} \\ y_{3t} \\ y_{4t} \end{bmatrix}$$

$$\underline{Y}_t = \begin{bmatrix} r_{1t} \\ r_{2t} \end{bmatrix} = \begin{bmatrix} 1 & R_{11t} & 0 & 0 \\ 0 & 0 & 1 & R_{21t} \end{bmatrix} \begin{bmatrix} d_{1t} \\ B_{1t} \\ d_{2t} \\ B_{2t} \end{bmatrix} + \begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{bmatrix}$$

$$\Rightarrow \underline{G}_t = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \underline{W}_t = \begin{bmatrix} y_{1t} \\ y_{2t} \\ y_{3t} \\ y_{4t} \end{bmatrix}, \quad \underline{W}_t = \begin{bmatrix} \sigma_{y_1}^2 & 0 & 0 & 0 \\ 0 & \sigma_{y_2}^2 & 0 & 0 \\ 0 & 0 & \sigma_{y_3}^2 & 0 \\ 0 & 0 & 0 & \sigma_{y_4}^2 \end{bmatrix}$$

$$V_t = \begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{bmatrix}, \quad \underline{V}_t = \begin{bmatrix} \sigma_{\varepsilon_1}^2 & r_{\varepsilon} \\ r_{\varepsilon} & \sigma_{\varepsilon_2}^2 \end{bmatrix}$$

$$\underline{F}_t = \begin{bmatrix} 1 & R_{11t} & 0 & 0 \\ 0 & 0 & 1 & R_{21t} \end{bmatrix}$$